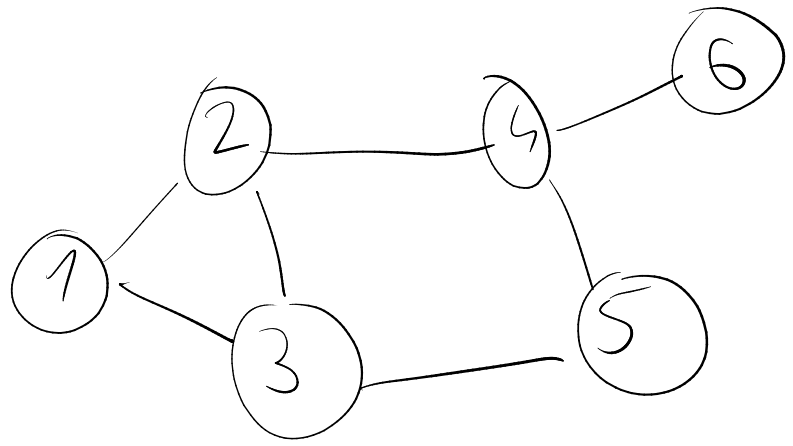


$\frac{1}{2}$ aproximativ

maximum matching

$G = (V, E)$

sol = $\emptyset$

while $E \neq \emptyset$

Alege (u, v)

$SOL = SOL \cup (u, v)$

$E = E \setminus E'$ (E' mulțimea muchiilor incidente cu u, v)

OPT - soluția optimă

SOL - algoritmul propus

$|SOL| \geq \frac{1}{2} |OPT|$

$OPT = \{(1,2), (3,5), (4,6)\}$

$SOL = \{(2,3), (4,5)\}$

Fie $e = (u,v) \in SOL$

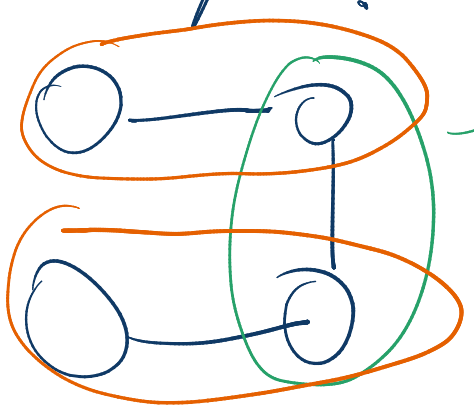
M_e - mulțimea muchiilor din OPT incidente cu $e = (u,v)$

$|M_e| \leq 2$

$$OPT \subseteq \bigcup_{e \in Sol} M_e \Rightarrow |OPT| \leq \sum_{e \in Sol} |M_e| \leq 2|Sol|$$

$$|SOL| \geq \frac{1}{2} |OPT|$$

Exemplu:

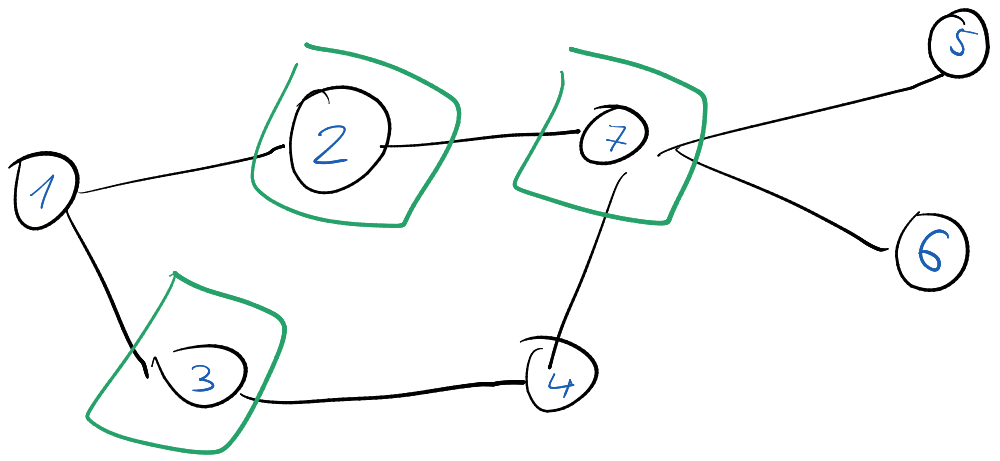


→ tight (pt alg nostru)

→ optim

Vertex cover

Trb să vedem toate muchiile din cât mai puține noduri.



2 aproximare

Folosim alg de mai devreme

Obs: vertex cover $\geq$ orice matching